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The development of a six-item short-form of the state scale of the Spielberger State–Trait Anxiety Inventory (STAI)

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Two studies are reported describing the development of a short-form of the state scale of the Spielberger State–Trait Anxiety Inventory (STAI) for use in circumstances where the full-form is inappropriate. Using item–remainder correlations, the most highly correlated anxiety-present and anxiety-absent items were combined, and correlated with scores obtained using the full-form of the STAI. Correlation coefficients greater than .90 were obtained using four and six items from the STAI. Acceptable reliability and validity were obtained using six items. The use of this six-item short-form produced scores similar to those obtained using the full-form. This was so for several groups of subjects manifesting a range of anxiety levels. This short-form of the STAI is therefore sensitive to fluctuations in state anxiety. When compared with the full-form of the STAI, the six-item version offers a briefer and just as acceptable scale for subjects while maintaining results that are comparable to those obtained using the full-form of the STAI.

The Spielberger State–Trait Anxiety Inventory (STAI) is one of the most frequently used measures of anxiety in applied psychology research. It is a reliable and sensitive measure of anxiety. Its popularity has meant that researchers are able to compare their results with those of others, a useful but rare occurrence in such research. The one drawback, however, with the STAI is its length, being 40 items long. This is a barrier to its use in studies where time for completing questionnaires is limited. It is also a barrier for some populations who are unable or unprepared to complete lengthy scales with seemingly repetitive items. These problems could be largely overcome by the development of a standardized short-form of the STAI.

The STAI consists of two questionnaires of 20 items each. The first questionnaire measures state anxiety (how one feels at the moment), the second, trait anxiety (how one generally feels). There are two STAI questionnaires: one produced in 1970 (Form X), the other in 1983 (Form Y). As yet there is no standardized short-form of the STAI. Some researchers (Foa, McNally & Murdock, 1989; McNally, Foa & Donnell, 1989; Smith, Ingram & Brehm, 1983) have used a 10-item STAI. The 10 items selected are alternate anxiety-present and anxiety-absent items from the full-form of the STAI. Where time is a consideration, Spielberger recommends using the 10-item anxiety score of the State–Trait Personality Inventory (STPI, 1979). None of these shortened forms of the STAI has been standardized.

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The aim of the two studies reported here is to develop a short-form of the state measure of the STAI for use in settings where time constraints preclude the use of the full-form.

STUDY 1: ITEM SELECTION

The purpose of this study is to select the smallest subset of both anxiety-present and anxiety-absent items from the full-form of the STAI which is highly correlated (i.e. $r > .90$) with scores obtained using the full-form of the STAI. Spielberger (1983) states that equal numbers of anxiety-present and anxiety-absent items constitute a more stable and consistent measure.

Method

Subjects

The sample comprised 200 pregnant women, from whom completed STAIs had been obtained as part of another study (Marteau, Johnston, Kidd, Michie, Cook, Slack & Shaw, 1992).

Procedure

The questionnaires were completed while women were waiting for a routine out-patient appointment. Scores on each item were correlated with scores on the remaining 19 items of the STAI (item-remainder correlations), to assess the contribution of each item to the overall state anxiety score.

Results

The items were ranked in order of magnitude of their item-remainder correlations (see Table 1). Equal numbers of anxiety-present and anxiety-absent items were then used to form two-, four-, six-, eight- and ten-item forms of the STAI. The correlations between scores from these short-forms and the full length STAI were: $r = .96$ for the 10 items; $r = .96$ for the eight items; $r = .95$ for the six items; $r = .91$ for the four items; and $r = .84$ for the two items.

Table 1. Item–remainder correlations: Pearson correlation coefficients between each item and the total of the remaining items of the STAI

Item no.	Adjective	*r*	Item no.	Adjective	*r*
1	*calm*	*.71*	11	self-confident	.50
2	secure	.66	12	nervous	.42
3	*tense*	*.62*	13	jittery	.48
4	regretful	.37	14	high-strung	.50
5	at ease	.66	15	*relaxed*	*.71*
6	*upset*	*.53*	16	*content*	*.69*
7	misfortunes	.51	17	*worried*	*.58*
8	rested	.42	18	over-excited	.33
9	anxious	.49	19	joyful	.40
10	comfortable	.58	20	pleasant	.63

Note. Italics depict the three highest anxiety-present and anxiety-absent items.

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Discussion

Correlations between subsets of four and six items of the STAI were greater than $r = .90$. Whether either is an acceptable short-form of the STAI will depend upon the extent to which they are reliable and valid in comparison with the full-form of the STAI.

STUDY 2: RELIABILITY AND VALIDITY

The aim of this study is to determine the reliability and validity of the four- and six-item short-forms of the STAI as a basis for choosing the optimal short-form. Internal reliability was assessed. Test–retest reliability was not calculated for, as Spielberger notes, test–retest reliability is low for state anxiety because of state anxiety's transitory nature (Spielberger, 1983). The form of validity for a short-form of the STAI to be assessed is concurrent validity, i.e. the extent to which scores on the short-form are similar to those obtained using the full-form. This is assessed for groups with mean STAI ratings near the population mean (unselected pregnant women), as well as groups with known raised anxiety (pregnant women following receipt of an abnormal test result during pregnancy).

Method

Subjects

The subjects comprised three groups: Group 1: 38 medical students, attending a psychology lecture; Group 2: 45 student nurses, attending a nursing lecture; Group 3: 200 pregnant women, as in Study 1, above; and Group 4: 23 pregnant women who had received an abnormal result on a routine screening test for fetal abnormality. Data from these women had been obtained as part of another study (Marteau, Cook, Kidd, Michie, Johnston, Slack & Shaw, in press).

Procedure

Questionnaires were completed while subjects waited for lectures (Groups 1 and 2), awaited a routine clinic appointment (Group 3), and at home, using a postal questionnaire (Group 4).

Results

The reliability coefficient for the 20-item STAI was: $\alpha = .91$; for the six items: $\alpha = .82$; for the four items: $\alpha = .77$.

Validity

The comparability of prorated scores from the six- and four-item scales with the use of the full scale was used to determine the concurrent validity of the two short-forms of the STAI. To avoid over-inflated correlations caused by items correlating with themselves (Pierce, Sarason & Sarason, 1986), the six-item questionnaire was compared with both the 20- and 14-item questionnaires, and the four-item questionnaire was compared with both the 16- and 20-item questionnaires. Comparisons were made using paired t tests (see Table 2).

There were no differences in the mean scores obtained using the full-form of the STAI and those obtained using the prorated six-item short-form, for any of the three

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Table 2. STAI scores (mean SD) derived from (*a*) the full-form, the prorated six-item short-form and the prorated 14-item short-form and (*b*) the full-form, the prorated four-item short-form and the prorated 16-item short-form

	Pregnant women ($N = 200$)	Student nurses ($N = 45$)	Medical students ($N = 38$)
(a)			
Full-form (20 items)	37.6 (9.1)	39.7 (7.0)	46.2 (6.2)
Prorated (6 items)	37.1 (11.0)	39.9 (7.5)	46.8 (6.2)
Prorated (14 items)	37.5 (8.7)	39.8 (6.9)	46.2 (7.1)
(b)			
Full-form (20 items)	37.6 (9.1)	39.7 (7.0)	46.2 (6.2)
Prorated (4 items)	38.4 (12.0)	41.4 (8.8)	48.6 (7.1)
Prorated (16 items)	36.9 (8.7)**	39.4 (7.0)*	45.6 (6.7)*

Note. Paired *t* tests were undertaken between data obtained from the prorated 6- (4-) item forms and both the full-form and prorated 14- (16-) item form.
* $p < .05$; ** $p < .01$.

groups. Similarly, there were no differences between the means prorated from the six-item short-form and the means prorated from the remaining 14 items. The prorated means obtained from the four item short-form were all significantly higher than the means based on the 16 remaining items and the full 20 items. Given this evidence suggesting the validity of the six-item short-form, further analyses were conducted with just this short-form of the STAI.

To determine whether the six-item short-form of the STAI is sufficiently sensitive to detect the changes in anxiety detected by the full-form, STAI scores obtained using the six-item short-form were compared with those obtained using the full-form on a sample with known elevated anxiety. Following receipt of an abnormal result on maternal serum alpha-fetoprotein screening indicating increased risk of spina bifida and Down's syndrome, women's anxiety scores are significantly raised compared both with their levels prior to testing, and when compared with women receiving a normal test result (Marteau *et al.*, in press). The mean STAI score for these women immediately following receipt of an abnormal test result is 46.4 (SD 14.8); the mean score prorated from the six-item short-form is 47.7 (SD 15.8): $t(22) = 0.96$, $p = .35$). Thus, the mean score obtained from the six-item short-form is similar to that obtained using the full-form of the STAI.

General discussion

The six-item short-form of the STAI (STAI-6: Appendix A) produces scores similar to those obtained using the full 20-item STAI. It has acceptable reliability, and produces scores that are similar to those produced with the full-form across subject groups manifesting normal and raised levels of anxiety. Two of the three anxiety-absent items are those identified by Spielberger to be particularly sensitive to low

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stressors; all three anxiety-present items he reports as being particularly sensitive to high stressors (Spielberger, 1983). When compared with the full-form of the STAI, the six-item short-form offers a briefer and more acceptable scale for subjects; it maintains results that are comparable to those obtained using the full-form of the STAI and remains sensitive to different degrees of anxiety.

There are several advantages to using a brief questionnaire. It is likely to maximize response rates, and minimize the number of response errors and unanswered items, thus improving the validity and generalizability of any findings. The longer a questionnaire, the more the responding sample will be biased in favour of those who find reading and writing easy. A brief questionnaire will also reduce the time taken to complete the questionnaire, and the time taken to score the results.

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Appendix A: Example from the Self-Evaluation Questionnaire (STAI-AD)

[Correction added on 20 December 2019 after first online publication: Appendix questionnaire has been updated.]

Directions

A number of statements which people have used to describe themselves are given on the following pages. Read each statement and then select the appropriate button to indicate how you feel ***right*** now, that is, ***at this moment***. There are no right or wrong answers. Do not spend too much time on any one statement but give the answer which seems to describe your present feelings best.

Not at all - Somewhat - Moderately so - Very much so



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